Preregistered extension: recurrence within an adversity domain and prediction from multiple prior events

Status. This registration covers a post-confirmatory extension of an existing UKHLS-HRS study. It was prepared after the parent study's exploratory and confirmatory phases, but before any coefficients or predictive-performance estimates for the extension were calculated.

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1. Background and scope

The parent study examines whether a mental-health response observed after one major adversity helps predict the response to a later adversity of a different kind. Exploratory work, a locally frozen confirmatory rebuild, and a separate measurement-adjudication analysis have already been completed. Those earlier analyses are not being preregistered retrospectively.

The present extension addresses two questions that remained after the parent analyses. First, is a response more reproducible when a broadly similar adversity recurs? Second, does a history based on at least two completed adversity episodes contain more useful information than a response observed after a single episode?

Before this registration, we conducted a coefficient-blind feasibility assessment. That assessment used event eligibility, missingness, timing, and sample counts only. It did not calculate response scores, correlations, regression coefficients, predictive R^2, RMSE, AIC, BIC, p values, or any other extension result. The feasibility files and their provenance will accompany the registration.

2. Feasibility information available before registration

2.1 Recurrence within the same adversity domain

In UKHLS, 2,078 participants are eligible for the acute-response recurrence analysis and 1,079 for the two-year-persistence analysis. These samples consist almost entirely of recurrent caregiving episodes: 1,990 participants for the acute outcome and 1,033 for persistence. Recurrent unemployment is uncommon (90 acute and 46 persistence pairs) and does not support a family-specific primary analysis.

HRS provides 221 acute and 171 persistence recurrence pairs in total. Because these samples do not reach the prespecified 300-person cohort threshold, HRS recurrence analyses are designated as secondary conceptual replications.

Recurrent health adversity and recurrent severe financial strain are retained as secondary analyses because the available recurrence definitions are less direct or more susceptible to reporting fluctuation.

2.2 Histories based on multiple prior events

For the acute outcome, UKHLS contains 1,603 participants who have at least two completed prior event responses, with at least one prior event from a domain different from the target event. Of these, 514 have historical response windows that share a mental-health observation. The primary UKHLS analysis therefore uses the 1,089 participants with non-overlapping historical windows; the full 1,603-person sample is a prespecified sensitivity analysis.

HRS contains 270 eligible participants for the acute multi-event analysis, including 82 with overlapping historical windows. HRS is secondary because it falls below the 300-person cohort threshold.

The corresponding samples for predicting two-year persistence or for jointly modelling acute response and persistence are too small for primary analysis (UKHLS: 162 and 152; HRS: 83 and 83). These analyses will not be promoted to primary status.

2.3 Direct same-domain versus cross-domain comparison

A strict comparison in which the same later target event has both a qualifying same-domain and a qualifying cross-domain history is not adequately powered. The available samples are 184 acute and 60 persistence cases in UKHLS, and 13 acute and 8 persistence cases in HRS. No confirmatory claim will be based on a direct head-to-head comparison of these small matched samples.

3. Research questions

Primary question 1: recurrence within caregiving

Does a person's response to an earlier caregiving episode improve out-of-sample prediction of the response to a later, clearly separated caregiving episode, beyond information about the later event and the person's mental health immediately before it?

We refer to this as a **within-domain recurrence benchmark**. It is not assumed to produce a positive result.

Primary question 2: prediction from multiple prior acute responses

Among people with at least two completed prior adversity episodes, does an aggregate of their prior acute responses improve out-of-sample prediction of the acute response to a later adversity, beyond later-event context and current pre-event mental health?

Secondary question: recency of mental-health information

On identical complete-case samples, how much predictive information is provided by an additional earlier pre-event mental-health measure once the most recent pre-event measure is known, and does event-response history add anything after both measures are included?

4. Data sources and analytical roles

UKHLS is the primary cohort. HRS is a secondary conceptual replication because its eligible recurrence and multi-event samples are smaller.

The outcome remains cohort-specific: GHQ-12 in UKHLS and the eight-item CES-D in HRS. Raw scores will not be pooled across cohorts. Existing event-episode files, event definitions, and censoring rules will be retained. No new event family will be introduced after registration.

5. Event definitions

5.1 Primary recurrence event

UKHLS caregiving recurrence. An episode begins when a participant moves from reporting no caregiving to reporting caregiving. For a later episode to count as a recurrence, at least one valid self-report wave after the earlier episode must show no caregiving, and the immediate pre-event wave for the later episode must also show no caregiving.

HRS parental-caregiving recurrence (secondary). An episode begins when a participant moves from reporting no personal care to a parent or parent-in-law to reporting such care. A later episode requires at least one valid no-care wave after the earlier episode and a no-care immediate pre-event wave.

5.2 Secondary recurrence events

The following are prespecified secondary analyses:

UKHLS unemployment recurrence after at least one valid wave in paid employment or self-employment;

HRS unemployment recurrence after at least one valid non-unemployed labour-force wave;

UKHLS recurrent persistent illness/disability after two consecutive valid waves without long-standing illness/disability;

HRS recurrent persistent ADL limitation after two consecutive ADL-free waves;

UKHLS recurrent severe financial strain after at least one valid non-severe-strain wave.

Repeated widowhood and repeated retirement are excluded.

6. Episode selection and temporal separation

For recurrence analyses, each participant contributes one pair: the earliest qualifying earlier episode and the first qualifying later recurrence. The earlier episode's two-year outcome must be observed before the later episode's immediate pre-event interview, and that interview must precede the later event interview. The two episodes may not share a transition or an outcome-observation window. Transitions containing more than one primary event are excluded.

For multi-event analyses, each participant contributes the earliest eligible target event that occurs after at least two completed prior isolated primary-event episodes. At least one prior episode must belong to a domain different from the target. Every item used to construct the historical predictor must have been observed before the target's immediate pre-event interview.

7. Outcomes

Within each cohort:

Acute response is the event score minus the immediate pre-event score.

Two-year persistence is the score closest to two years after the event minus the immediate pre-event score.

Primary analyses use cohort-standardized outcomes. Standardization parameters will be estimated within each outer training set and then applied to its test set. Raw-scale analyses are prespecified sensitivities.

The recurrence benchmark evaluates both acute response and two-year persistence. The multi-event primary analysis evaluates acute response only, because the persistence and joint samples do not meet the primary sample-size threshold.

8. Predictors

8.1 Context variables

Target-event family, age at the target pre-event interview, sex, education, target calendar year, and the number of previously observed primary events.

8.2 Current state

The mental-health score measured immediately before the target event.

8.3 Historical response for the recurrence analysis

The corresponding response to the earlier episode in the same domain: earlier acute response when predicting later acute response, and earlier two-year persistence when predicting later persistence.

8.4 Multi-event historical representations

The primary historical predictor is the arithmetic mean of all qualifying prior acute responses. The number of prior episodes is retained as a context variable.

Prespecified secondary representations are:

the most recent qualifying prior acute response;

the mean prior acute response after training-fold residualization for prior-event family, prior pre-event mental health, age, calendar year, and number of preceding events;

the between-history standard deviation among participants with at least two qualifying histories.

We will not add latent classes, empirical-Bayes scores, random-slope models, or unrestricted machine-learning feature searches after registration.

9. Model comparisons

All comparisons are prospective and evaluated on held-out participants.

9.1 Recurrence within caregiving

R0: context variables only.

R1: R0 plus the earlier same-domain response.

R2: R0 plus current pre-event mental health.

R3: R2 plus the earlier same-domain response.

The primary increments are R1 versus R0 and R3 versus R2.

9.2 Multi-event acute-response history

M0: context variables only.

M1: M0 plus the aggregate prior acute-response history.

M2: M0 plus current pre-event mental health.

M3: M2 plus the aggregate prior acute-response history.

The primary increments are M1 versus M0 and M3 versus M2.

9.3 ANCOVA sensitivity

For each analysis, an ANCOVA formulation predicts the later event-level outcome and includes the later pre-event score as a mandatory covariate. The ANCOVA and change-score results will be reported together; neither will be selected on the basis of performance.

10. Validation and uncertainty

The primary validation uses deterministic 10-fold cross-validation grouped by participant (seed 20260904). All episodes and histories from a participant remain in the same fold. Standardization, preprocessing, and residualization are fitted in the outer training data only.

A temporal holdout will train on the earlier 70% of eligible target-event calendar time and test on the later 30%. Participants present in the training period will be excluded from the temporal test set.

The primary performance measure is incremental predictive R^2 on held-out data. Predictive R^2 is defined as 1 - SSE/SST_test, and increments are calculated between nested models on the same test observations. RMSE, MAE, mean log predictive density, and calibration intercept and slope are secondary measures.

Uncertainty will be summarized with 500 successful paired bootstrap resamples at the participant level. All records belonging to a participant will be resampled together.

11. Overlapping historical windows

The primary multi-event analysis excludes participants whose qualifying historical response windows share a mental-health observation. The broader sample that permits such overlap is retained as a prespecified sensitivity analysis. This distinction is intended to prevent the same psychological measurement from contributing to more than one historical response and thereby inflating apparent stability.

12. Secondary analyses

The following analyses are secondary regardless of their results:

HRS recurrence and multi-event analyses;

recurrent unemployment in UKHLS and HRS;

recurrent health adversity in UKHLS and HRS;

recurrent severe financial strain in UKHLS;

raw-scale outcome analyses;

the recency comparison using fixed pre2 and pre1 measures on identical complete-case samples;

the multi-event analysis that permits overlapping historical windows.

Secondary analyses will not replace a null or inconclusive primary analysis.

13. Analyses outside the confirmatory extension

The following will not be treated as confirmatory analyses:

a direct same-domain versus cross-domain comparison on a common target event, because the matched samples are too small;

multi-event prediction of later two-year persistence or a joint acute-plus-persistence target, because neither cohort reaches the 300-person primary threshold;

subgroup searches by age, sex, education, income, or event severity;

post-result changes to event thresholds, event definitions, or historical features.

14. Interpretation rules

A positive recurrence increment together with little cross-domain transfer will be interpreted as evidence that response information may generalize more readily within a recurring caregiving context than across different adversities.

A negligible recurrence increment will be interpreted as evidence that a single episode response has limited predictive stability even when the broad domain recurs.

For the multi-event analysis, an increment above 0.0025 after current-state adjustment, with a UKHLS bootstrap interval excluding zero and an HRS estimate in the same direction, will be interpreted as evidence that aggregation across several prior acute responses recovers limited transportable information. Otherwise, we will conclude that the observed multi-event acute history does not materially improve on current-state assessment under the registered prediction design.

The descriptive incremental predictive-R^2 reference values are 0.0025, 0.005, and 0.010. They are interpretive benchmarks, not clinical thresholds. No result will be described as proving or disproving the existence of resilience, a stable trait, or a causal mechanism.

15. Reporting and provenance

We will archive eligibility files, fold assignments, out-of-fold predictions, bootstrap summaries, source and code hashes, and software versions. Reports will distinguish this externally registered extension from the parent study's exploratory and locally frozen confirmatory analyses. The registration will state explicitly that the extension was developed after the parent analyses but before any extension response coefficient or predictive-performance estimate was viewed.